SYSTEM ARCHITECTURE

Semiconductor Knowledge Graph

End-to-end data pipeline from raw tables to artificial intelligence.

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Modular Pipeline Stages

Data cleaning through advanced link prediction.

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Parts 01 & 02: Foundation

Part 01: Profiling & Cleaning

- OpenRefine manual data correction.
- Flipping import logic for mirrored trade networks.
- Strict rule checks (null values, negatives, self-trade).

Part 02: Treatment

- CSV data aggregation and mapping.
- Total exact trade value calculation for accuracy.

Raw CSV Schemas

Trade Data	year, tier, exporter_iso, exporter_name, importer_iso, importer_name, trade_value_usd
Country Profiles	country_iso, country_name, total_export_usd, total_import_usd, net_trade_balance_usd, strategic_role
Supply Tiers	hs_code, tier_id, category_name, strategic_bottleneck

Part 03: RDF Materialization

N-Triples Generation

Automated Python declarative mapping from minimized CSVs to strict node URIs and typed literals.

Explicit Tier Linking

Manual semantic definition of supply chain flows (e.g., Wafers feeding into Logic Processors).

Apache Fuseki

Live triplestore deployment target for raw materialized graphs and external query endpoints.

Parts 04 & 05: Logic and Validation

Part 04: OWL Ontology

TBox Rules
Mathematical logic injected into physical data.

Transitivity
Enabling multi-tier supply chain inferencing.

Strict Disjointness
Immutable separation of Countries, Tiers, and Observations.

Part 05: SHACL & Provenance

Acyclic Supply Chain Verification
SPARQL-based checks preventing logical loops.

Graceful Degradation
Separating hard Violations from real-world data Warnings.

PROV-O Immutability
SHA-256 hashes locking exact execution audit trails.

Part 06 & 07: Deployment: AI Integration

LLM Access Layer

Gemini-powered natural language to SPARQL translation with self-healing syntax validation.

Graph Machine Learning

TransE link prediction (PyTorch/PyKEEN) discovering missing Exporter-Tier-Importer triads.

Docker Orchestration

Single root container architecture bridging ephemeral Python environments with persistent Fuseki volumes.

Live Demo: Docker Web Interface